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MAJOR PROJECT PRESENTATION

Earth Air Heat Exchanger for building application

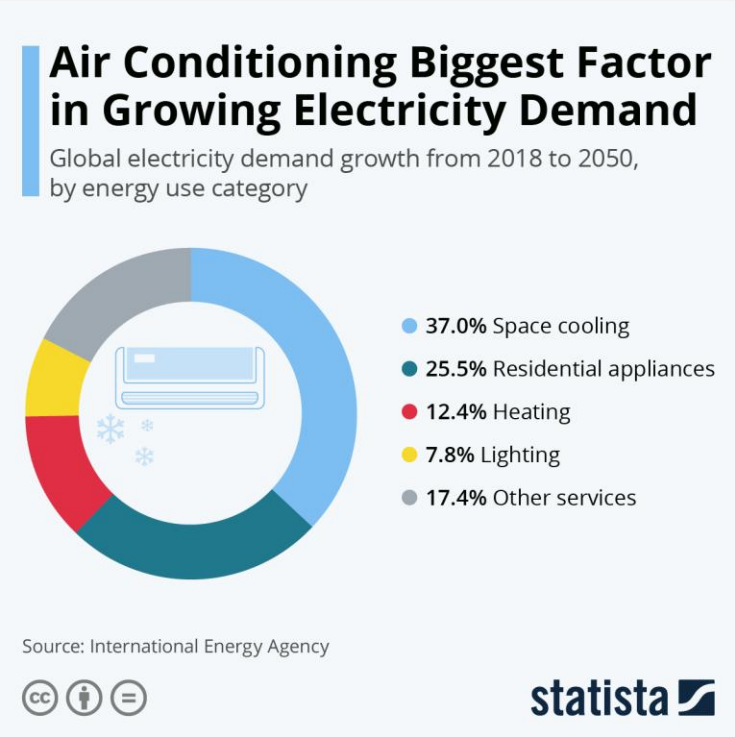
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INTRODUCTION



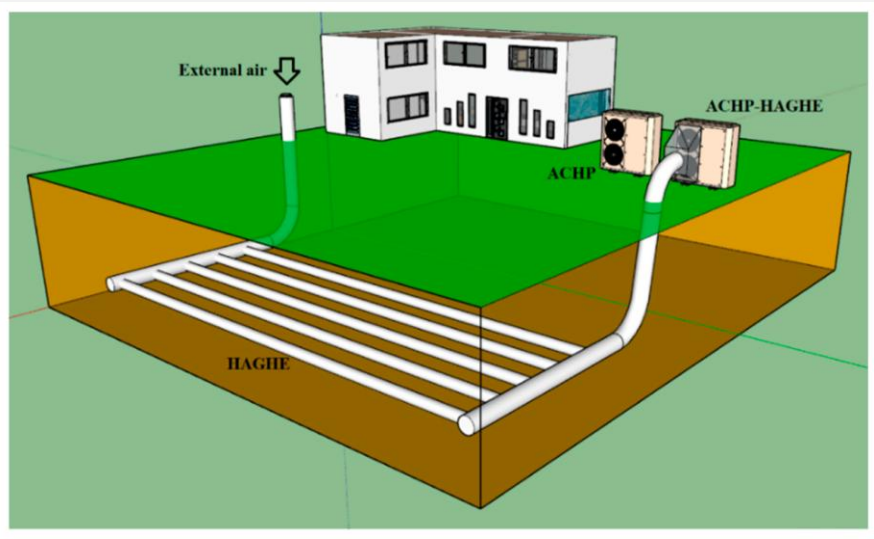
According to a report by the Ministry of Statistics and programme implementation, the annual electricity consumption per capita in India stands at 1015 kWh, which corresponds to approximately two tons of carbon emissions per capita if the electricity were produced from coal.

Rising temperature and frequent heat waves in India have intensified electricity demands, driving primarily by air conditioners.

The International Energy Agency (IEA) estimates that by 2050, space cooling will account for 45% of the country's electricity demand.

Hence, there is a need for innovative solutions that integrates renewable sources of energy such as solar energy, wing energy and geothermal energy within the HVAC systems to reduce its load requirements and therefore, reducing carbon emissions.

SOLUTION: EAHE



Earth Air Heat Exchanger (EAHE) is designed to utilize the Earth's stable underground temperature to preheat or precool the air which can reduce the load of heaters during winter and air conditioners during summers.

At a depth of 1.5 m to 3 m under the ground, the ground maintains a relatively stable temperature throughout the year. This temperature remains cooler than ambient air during summers and warmer than ambient air during winters. As a result, when the ambient air passes through a pipe at this depth, heat transfer takes place due to the temperature difference between ambient air and the constant soil temperature

LITERATURE REVIEW

Evaluation of Earth-Air Heat Exchanger (EAHE) Systems in Various Applications

Experimental Investigation of EAHE in Hot Arid Climates

- Experimental Setup
- g Capacity Results

EAHE for Space Heating and Cooling in Egypt

- Experimental Setup
- Extreme Conditions

EAHE Cooling Potential in Agricultural Greenhouses in Thailand

- p and Configuration
- on Cooling Efficiency

EAHE Projects in Office Buildings in Germany

- Impact on Indoor Tempe
- Overview of the Investigated
- Energy Gains from EAHE Syst

Integration of EAHE in a Solar Greenhouse in Turkey

- System Design and C
- Performance Metrics

LITERATURE REVIEW

Evaluation of Earth-Air Heat Exchanger (EAHE) Systems in Various Applications

Laboratory Simulator for EAHE Efficiency

- Study Overview and Methodology
  - This investigation employed a laboratory EAHE simulator, replicating real ground conditions with an 8.7 m long, 101.8 mm diameter PVC pipe.
  - An artificial neural network model was integrated to predict outlet air temperatures.
- Findings on Efficiency and Temperature Reduction
  - The study reported efficiency rates as high as 88%, with an approximate 10°C reduction in air temperature.
  - These results highlight the potential of data-driven control strategies for optimizing EAHE performance.

Experimental Investigation of EAHE in Hot Arid Climates

- Overview of the Experimental Setup
  - An experimental setup in a hot, arid region featured a compact EAHE configuration with a 1.5 m long, 0.15 m diameter pipe buried in a galvanized steel drum.
- Cooling Capacity Results
  - The study showed that the system could reduce the outlet air temperature by up to 22°C.
  - These results underline the robust cooling capacity of EAHE systems in extreme conditions.

EAHE for Space Heating and Cooling in Egypt

- System Design and Experimental Setup
  - The study involved a 5.5 m long horizontal EAHE system made of PVC, buried in loamy sand soil at a depth of 2 m.
  - Air circulation was maintained at speeds between 1-3.9 m/s.
- Performance in Extreme Conditions
  - Measurements taken during both summer and winter revealed that the outlet air temperature stabilized around 25°C.
  - This performance was noted even when ambient conditions exceeded 40°C, confirming the system's potential for maintaining indoor comfort in arid climates.

Trombe e

EAHE Im Guest Ho

EAHE Pro Buildings

EAHE Cooling Potential in Agricultural Greenhouses in Thailand

Experimental Setup and Configuration

This investigation assessed the cooling potential of an EAHE system in an agricultural greenhouse.

The system utilized an iron tube arranged in six serpentine rows, with PVC inlet and outlet tubes of 0.08 m diameter, buried at a depth of 1 m.

Results on Cooling Efficiency

Results demonstrated significant cooling efficiency, making the system particularly suitable for tropical climates.

The findings underscore the adaptability of EAHE systems in various agricultural applications.

EAHE Performance in a School Building in Italy

Study Overview and Setup

This study monitored an extensive EAHE installation in a school in Imola, Italy, featuring three sets of buried pipes.

Each set consisted of pipes 70 m in length and 0.25 m in diameter, with an average burial depth of 2.61 m.

Impact on Indoor Temperature and Comfort

Continuous monitoring over a 12-month period revealed that the EAHE system maintained stable indoor temperatures.

The system effectively moderated both heating and cooling loads, improving overall thermal comfort in the school environment.

Trombe et al. Study in France

- Overview of the Experimental Setup
  - Conducted in Toulouse, France, comparing two similar residential buildings: one with an EAHE system and one without.
  - Utilized PVC tubes with a diameter of 0.2 m and thickness of 5 mm, buried at a depth of 2.5 m with a length of 42 m.
  - Adjustable airflow rate ranged from 0 to 45 m^3/h, allowing for controlled experimentation.
- Findings on Temperature and Humidity
  - Hourly measurements indicated a wide variation in inlet air temperature (18-36°C).
  - The outlet temperature remained significantly stabilized between 18-25°C, demonstrating the effectiveness of the EAHE system in temperature regulation.

EAHE Implementation in a Guest House in India

- System Configuration and Setup
  - An EAHE system was installed in eight rooms of a double-floor guest house.
  - Multiple concrete pipes were used, each 85 m long and 0.5 m in diameter, buried at a depth of 2.5 m with a mean air velocity of 6.3 m/s.
- Impact on Indoor Temperature
  - Data recorded every two hours over a month showed that air-conditioned rooms maintained indoor temperatures 2-5°C lower than non-air-conditioned rooms.
  - Ambient temperature fluctuations ranged between 22.5 and 44.2°C, highlighting the system's effectiveness in maintaining comfort.

Involved three EAHE projects across various office buildings with significant

Integration of EAHE in a Solar Greenhouse in Turkey

System Design and Configuration

The study integrated an EAHE system with a solar greenhouse in Izmir, featuring a horizontal U-bend EAHE.

The system included a galvanized pipe 47 m long and 0.56 m in diameter, buried at a depth of 3 m, along with an auxiliary tube of 15 m length and 0.8 m diameter.

Performance Metrics and Efficiency

Extensive temperature and humidity measurements confirmed an exergy efficiency of approximately 60.7%.

The results indicate the potential of hybrid EAHE applications in enhancing greenhouse performance.

EAHE System in a Residential Building in Brazil

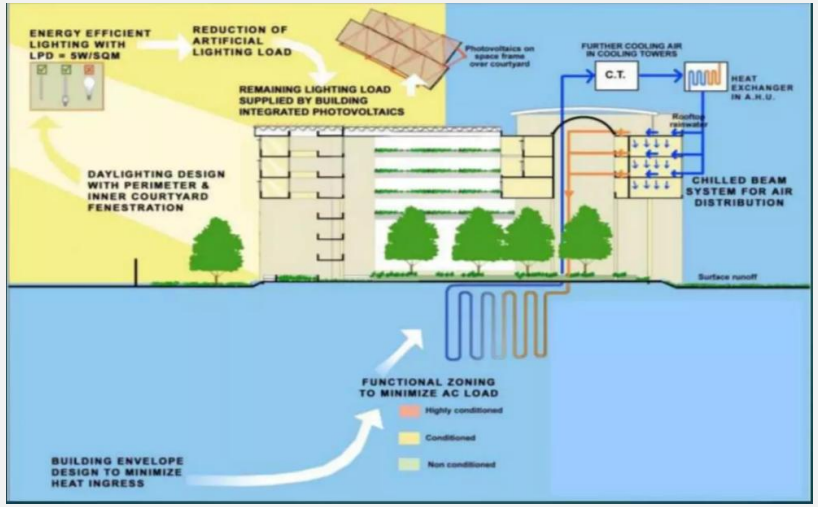
Configuration of the EAHE System

The EAHE system comprised three buried PVC tubes, with two tubes having a diameter of 0.11 m and a burial depth of 2 m, and one tube with a diameter of 0.10 m and a burial depth of 0.5 m.

Results on Indoor Thermal Comfort

Experimental results indicated that for tubes buried at 2 m, the internal air temperature could increase by over 8°C or decrease by about 4°C.

These findings highlight the system's robust heating and cooling potential, enhancing indoor thermal comfort.



EAHX IMPLEMENTATION WORLDWIDE

Indira Paryavaran Bhavan, New Delhi

- Type: Vertical closed-loop EAHX
- 180 vertical bores with 32mm outer diameter and 3m apart
- Bore Depth: 80 meters
- Energy Savings: 40% lower HVAC load
- Outcome:
  - India's first net-zero energy building

Greenhouse & Animal Housing, USA

- Type: Horizontal pipe 2-3 meters depth
- Pipe Material: PVC and HDPE
- Temperature Regulation: Pre-cools in summer and Preheats in winter
- Outcome:
  - Reduced heating and cooling costs for agricultural production

Passive houses in Europe: Austria, Denmark, Germany

- Pipe Depth: 1.5 - 2 meters below ground : Horizontal type
- Pipe Length: 30-50 meters (varies by design)
- Airflow Rate: 100 - 500 m^3/h
- Efficiency: Saves 30-50% on heating/cooling costs
- Outcome:
  - Improves indoor air quality while reducing energy usage

AIM



- To validate the theoretical modeling of EAHE with the experimental model.
- To calculate the total reduction in cooling load of air conditioners during summer season
- Calculate the total embodied energy needed to construct the system, energy payback time (EPBT), total CO_2 emission, CO_2 mitigation and carbon credit earned.

ASSUMPTIONS:

The following assumptions are considered for thermal modelling of EAHE.

1. The soil is isotropic meaning that the soil properties do not change for different directions. This means that its conductivity will remain constant throughout the depth.
2. Because of the pipe’s extremely thin thickness, it has very little heat resistance. It means that the pipe's surface temperature is equivalent to the soil temperature.
3. The temperature of the soil is the yearly average temperature of New Delhi which is 25°C and is constant throughout the depth.
4. The PVC pipe used is of uniform cross-sectional area and has a smooth inner surface ensuring no slip condition.

From: Cengel, Yunus A., and Afshin J. Ghajar. Heat and Mass Transfer: Fundamentals and Applications.

THERMAL MODELLING FOR EAHE

1. Cross-sectional area is given by

$$A_c = \frac{\pi D^2}{4}$$

2. The curved surface area of the pipe where the heat transfer would take place is given by

$$A_s = \pi \mathrm{DL}$$

3. The mass flow rate of air is given by

$$\dot{m} = \frac{\rho v A_c}{N}$$

Where $density\ of\ air\ at\ 27℃, \rho = 1.204 \frac{kg}{m^3}$

$specific\ heat\ of\ air\ at\ 27℃, c_p = 717.1\ kJ/kgK$

4. Convective heat transfer coefficient inside the pipe is given by

$$h = \frac{N_u K}{D_h}$$

where $N_u\ is\ the\ nusselt's\ number$

K is the thermal conductivity, K = 0.025 W/m-K

D_h is the hydraulic length which in a circular tube is equal to the diameter of the pipe.

5. Nusselt's number: (turbulent flow)

$for\ a\ smooth\ and\ circular\ tube\ it\ is\ given\ by\ Gnielinski\ (1976)$

$$Nu = \frac{\left(\frac{f}{8}\right)(Re-1000)Pr}{1+12.7\left(\frac{f}{8}\right)^{0.5}\left(Pr^{\frac{2}{3}}-1\right)}$$

$0.5 \leq \mathrm{Pr} \leq 2000$

$3 \times 10^3 \leq \mathrm{Re} \leq 5 \times 10^6$

Where f is the friction factor

Pr is the Prandtl number

Re is the Reynolds number

6. Friction factor, for a smooth pipe is given by Petukhov (1970) equation

$$f = (0.790 \ln(Re) - 1.64)^{-2}$$

$3 \times 10^3 \leq \mathrm{Re} \leq 5 \times 10^6$

7. Prandtl Number is given by :

$$\mathrm{Pr} = \frac{\rho V c_p}{K}$$

Where cp is the specific heat of air at 27℃, cp = 717.1 J/kg-K

8. NTU (number of transfer units) is given by:

$$NTU = \frac{hA_s}{\dot{m}c_p}$$

9. Now, the Exit temperature can be determined from the equation:

$$T_e = T_s - (T_s - T_i)e^{-NTU}$$

Where Ts and Ti are the surface and inlet temperature of air, respectively.

10. Rate of heat transfer in the EAHE:

$$T_l = \frac{T_i - T_e}{log(\,(T_s - T_e)\,/\,(T_s - T_i))}$$

$$Q_u = h \times A_s \times T_l$$

THERMAL MODELLING FOR EAHE+ROOM

3 Unknowns:

T_r

T_{win}

T_{Wout}

1. Absorbed Radiation Energy:

$$\alpha\left(\frac{I_{total}}{3600}\right)A = \frac{kA(T_{win} - T_{wout})}{L} + AU_o(T_{wout} - T_a)$$

Conduction

Losses

2. Conduction through wall:

$$\frac{kA(T_{win} - T_{wout})}{L} = hA(T_r - T_{win})$$

Convection

3. Convection inside the room:

$$hA(T_r - T_{win}) + Q_u = Mc_p\frac{dT_r}{dt} + \rho VNc_p(T_r - T_a) + UA(T_r - T_a)$$

Rate of change of thermal energy (energy stored in the system)

Losses

Energy input from EAHX

Heat transfer due to infiltration of air

REDUCTION IN COOLING LOAD

- After the theoretical model is validated, the cooling load calculation is considered for a realistic size of room (12 ft × 12ft × 12 ft)
- Summer season is considered of four months (March, April, May and June).
- Cooling load reduction is calculated for a single day (4 April) and has been extrapolated to 4 months i.e., 122 days.
- This is good assumption since the lower temperature difference between ambient and soil in March is compensated by higher temperature difference in May.

Total cooling load of room for air conditioner:

$$Q_{AC} = \rho V(N \times 3600)c_p(T_a - 24)$$

Reduction in cooling load:

$$Q_{red} = \rho V(N \times 3600)c_p(T_a - T_r)$$

CARBON CREDIT EARNED

Annual CO_2 emission per year = $\underbrace{\frac{E_{in}}{L} \times \frac{1}{1-L_a} \times \frac{1}{1-L_t} \times 0.98\ kg}_{2.04}$

- Transmission losses (L_t) = 0.4 and domestic appliances losses (L_a) = 0.2
- The average CO_2 equivalent intensity for electricity generation from coal is approximately 0.98 kg of CO_2 per kW h at the source.

The net CO_2 mitigation over the lifetime of the system = Total CO_2 mitigation - Total CO_2 emission

Net CO_2 mitigation = $(E_{aout} \times L\ - E_{in}) \times 2.04\ kg$

Carbon Credit Earned = $Net\ CO_2\ mitigation\ \times 10\ \times 84.67$

EXPERIMENTAL SETUP



Figure 1 Photos of the constructed EAHE system.

a. The complete EAHE system. b. The PVC pipe buried at 6 feet depth. c. Data Logger. d. Plastic blower of 550 W.

Construction Detail:

- Cement Cubical Room: 4 × 4 × 4 feet
- Pipe material : PVC
- Pipe diameter: 0.3m
- Horizontal Pipe buried @ 6 feet deep in horizontal configuration
- Effective Pipe Length: 15 feet
- Plastic Blower Fan: 550W

Experimental Details:

- Done on 22 March, 3 April and 4 April with air flow velocity as 0.5, 1.0 and 1.5 m/s respectively.

EXPERIMENTAL DATA

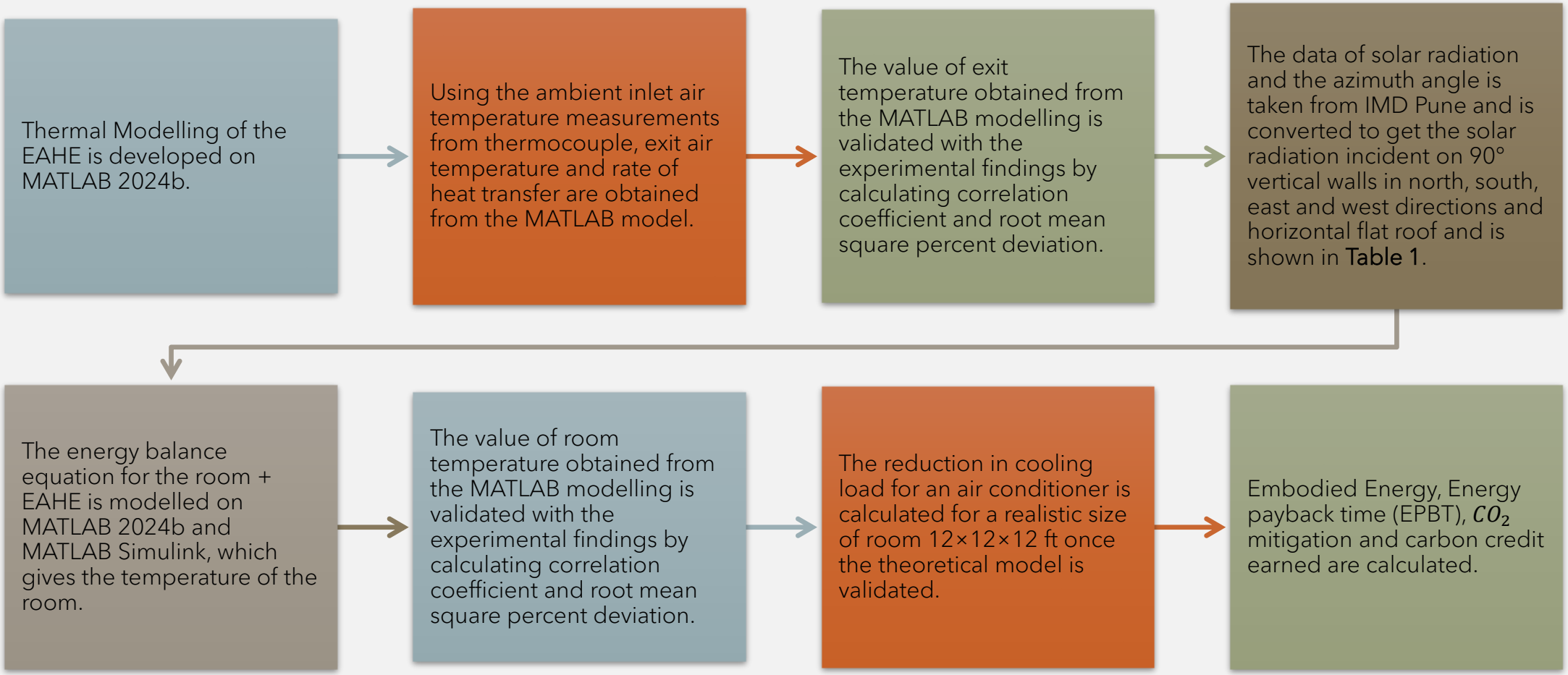
To measure the temperatures of air at inlet and outlet of the heat exchanger as well as to measure the wall temperatures of the room, **K type thermocouples** are used.

These thermocouples are connected to a **data logger** to save the values of the temperature every 5 min interval.

A **mercury thermometer** was used to measure the ambient temperature and room temperature every hour.

A **digital anemometer** is used to measure the velocity of air at the exit of the heat exchanger.

METHODOLOGY



ANALYSIS PARAMETER

The validation of the experimental results with theoretical results are done by evaluating the correlation coefficient (*r*) and the root mean square percent deviation (*e).* The root mean square percent deviation indicates the percent deviation of experimental results from the theoretical results.

These are given by the following equations:

Correlation coefficient r = $\frac{N\sum X_iY_i - \sum X_i)(\sum Y_i)}{\sqrt{N\sum X_i^2 - (\sum X_i)^2}\sqrt{N\sum Y_i^2 - (\sum Y_i)^2}}$,

Where X_i is theoretical values, Y_i is experimental values

Root Mean Square Percent deviation, $e = \sqrt{\frac{\sum (e_i)^2}{N}}$,

Where $e_i = \frac{X_i - Y_i}{X_i} \times 100$

For theoretical results to have a strong correlation with the experimental results, the correlation coefficient should be between 0.8 and 1.

And RMSE value should be less than 5%.

SOLAR RADIATION INCIDENT ON DIFFERENT WALLS

Day	Time	North Wall (W/m^2h)	South Wall (W/m^2h)	East Wall (W/m^2h)	West Wall (W/m^2h)	Roof Wall (W/m^2h)	**Total Radiation** (W/m^2h)
22 March	10:00 AM	453.52	149.03	149.03	395.60	677.83	1825.00
	11:00 AM	723.50	142.28	152.43	142.28	808.40	1968.89
	12:00 PM	793.87	140.93	140.93	309.79	875.07	2260.59
	1:00 PM	765.64	154.51	312.56	154.51	877.85	2265.06
	2:00 PM	540.36	177.73	472.43	177.73	808.40	2176.66
	3:00 PM	337.79	193.90	408.03	193.90	669.50	1803.12
	4:00 PM	221.34	186.05	273.39	186.05	488.93	1355.76
	5:00 PM	134.20	131.07	144.62	131.07	269.47	810.41
3 April	10:00 AM	446.71	151.97	151.97	531.94	755.62	2038.21
	11:00 AM	777.83	137.76	137.76	383.46	875.07	2311.87
	12:00 PM	912.86	133.43	215.35	133.43	936.19	2331.25
	1:00 PM	829.68	146.94	409.02	146.94	933.41	2466.00
	2:00 PM	722.90	174.32	231.98	174.32	866.74	2170.25
	3:00 PM	396.51	199.98	453.35	199.98	738.95	1988.78
	4:00 PM	259.33	209.73	332.50	209.73	580.60	1591.88
	5:00 PM	183.46	159.47	163.27	159.47	341.69	1007.37
4 April	10:00 AM	447.73	150.90	150.90	533.57	755.62	2038.73
	11:00 AM	780.63	136.54	136.54	383.79	875.07	2312.58
	12:00 PM	916.27	132.14	214.55	132.14	936.19	2331.28
	1:00 PM	832.67	145.65	409.37	145.65	933.41	2466.75
	2:00 PM	725.57	173.09	231.16	173.09	866.74	2169.66
	3:00 PM	397.18	198.91	454.52	198.91	738.95	1988.46
	4:00 PM	259.08	208.85	333.18	208.85	580.60	1590.56
	5:00 PM	183.49	158.94	162.83	158.94	341.69	1005.88

AMBIENT TEMPERATURE

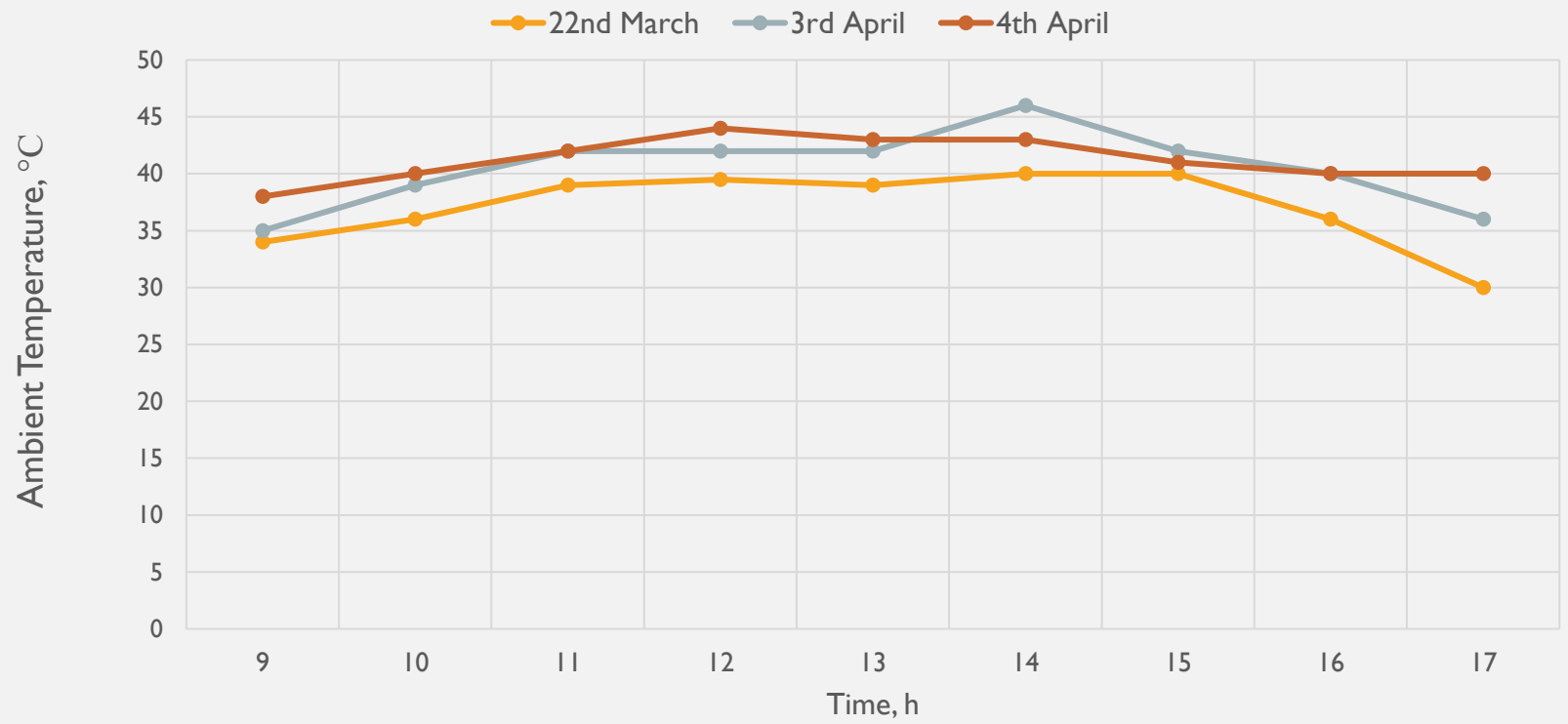


Figure 3 The hourly variation of ambient temperature on the experimental days.



RESULTS AND DISCUSSION

EXPERIMENTAL OBSERVATION

- On 22nd March, it is observed that the inlet temperature ranges from 32.2°C to 41.7°C.
- And the temperature difference ranges from 1.9 to as high as 7.3.
- This difference corresponds to the lowest inlet temperature of 32.2°C and 41.2°C respectively which is close to the highest temperature.
- Similar observations were made on 3rd April and 4th April.
- Hence, it is correct to say that when the temperature difference between inlet and the soil temperature (which in this case is 25°C), is more then, the temperature difference between inlet and exit is more which corresponds to more rate of heat transfer.

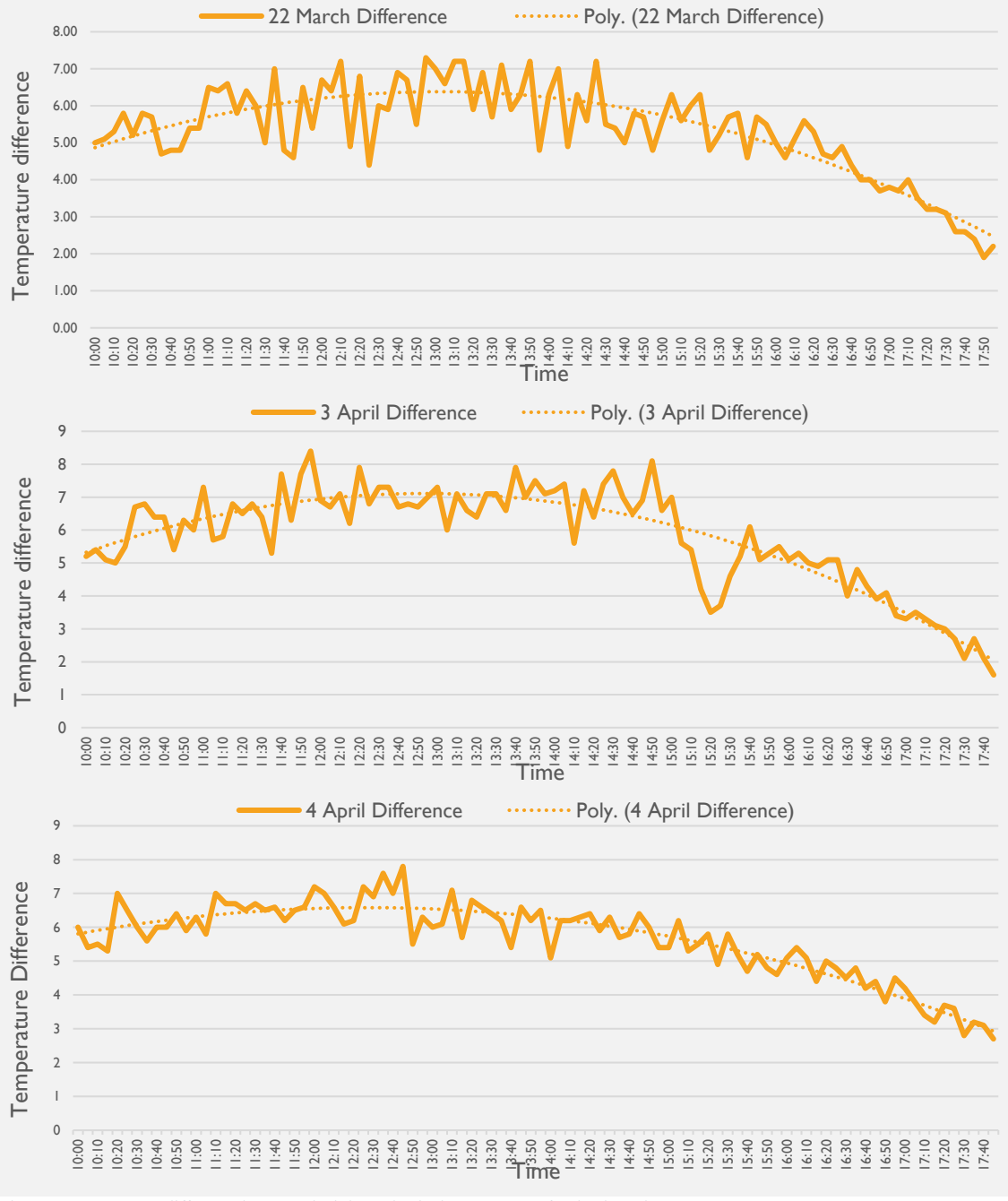
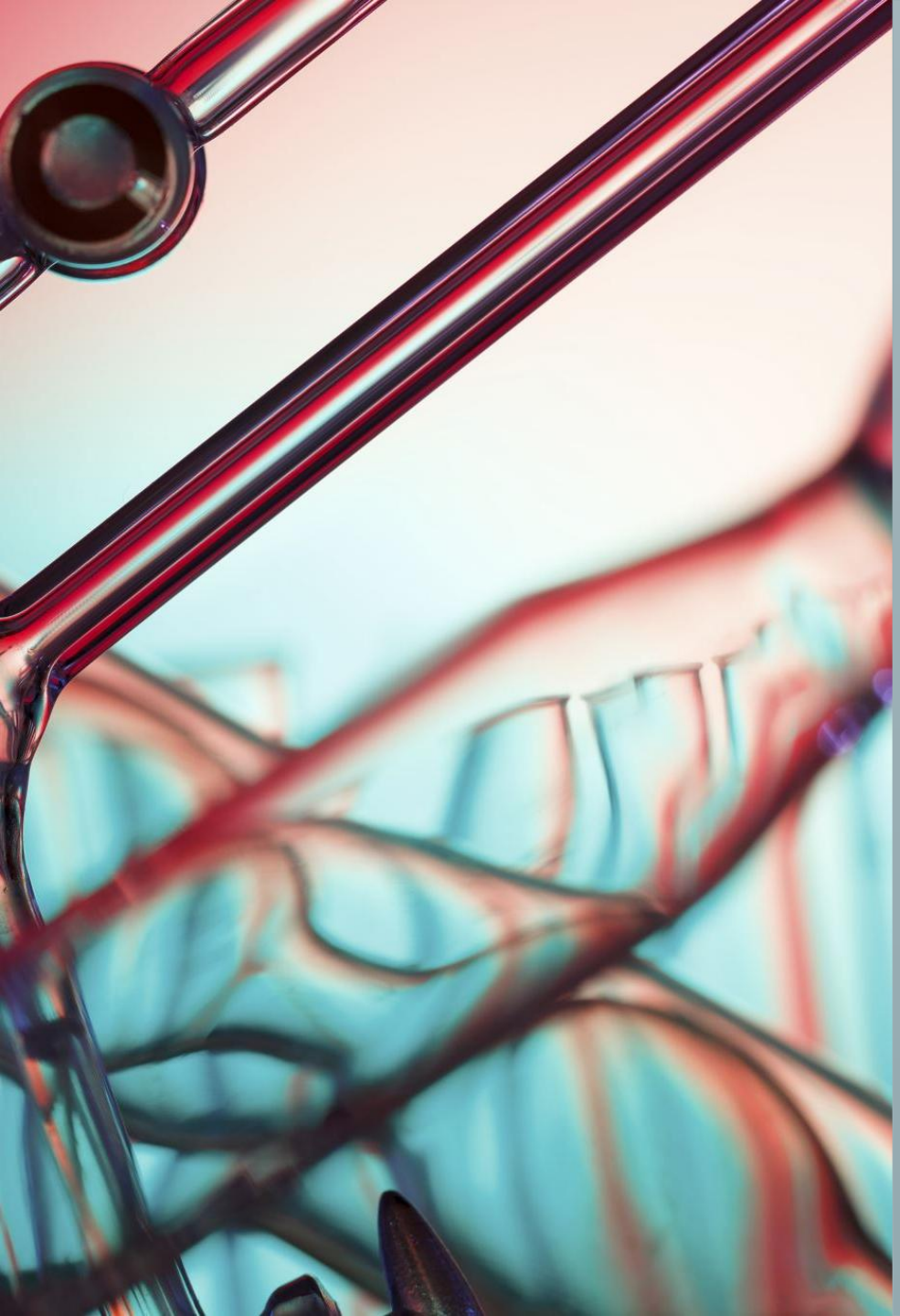


Figure 2. Temperature difference between the inlet and exit air temperature for the three days



EXPERIMENTAL VALIDATION

VALIDATION OF EAHE THERMAL MODELLING

- It can be observed that the theoretical model is in good agreement with the experimental obtained values.
- The correlation coefficients are between 0.84 to 0.89 and the RMS percent deviation is less than 3%, which shows the percent deviation is minimal.
- Hence, the theoretical model results are validated.

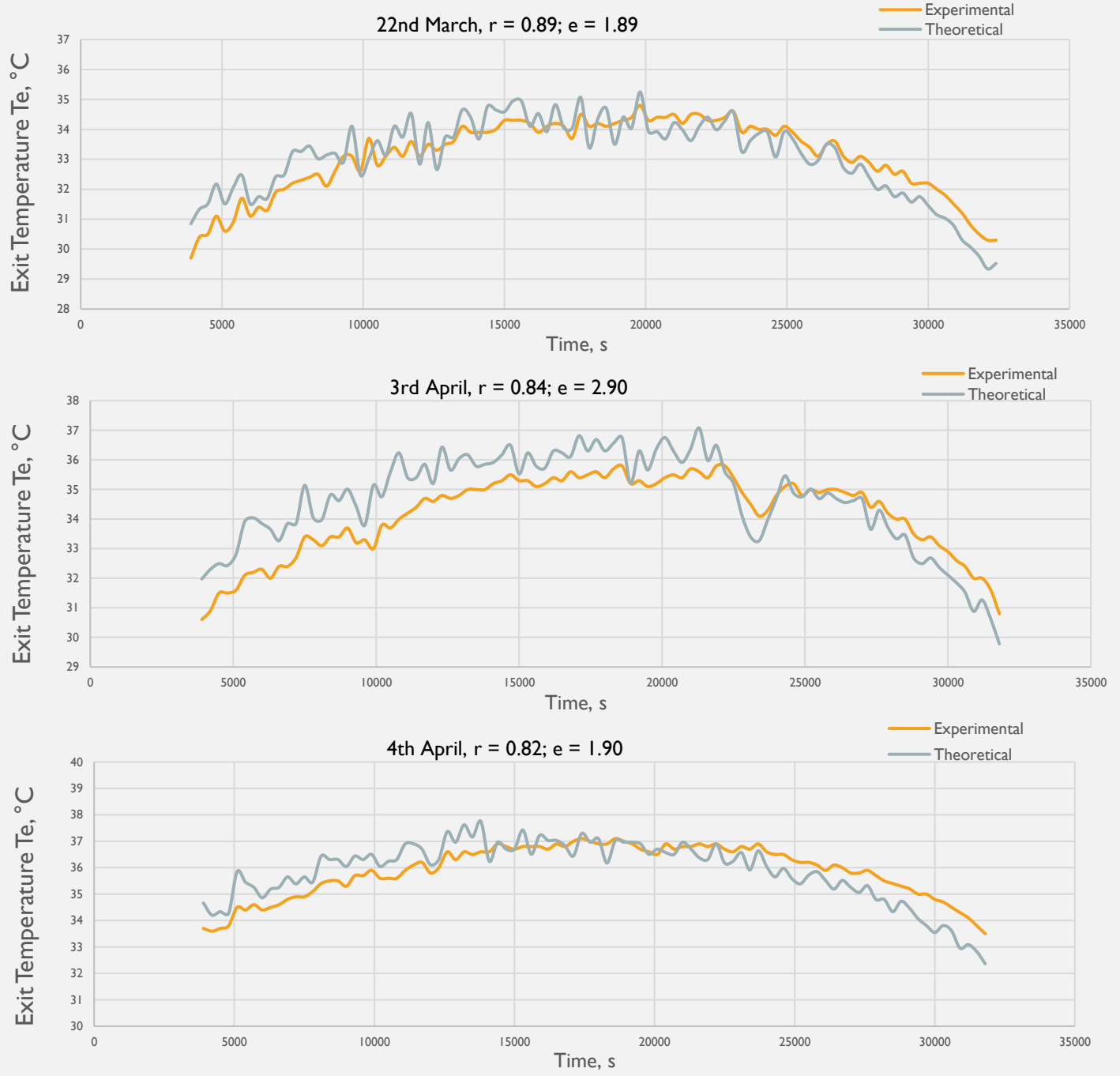


Figure 4 Variation of theoretical exit temperature with experimental validation for each day.

VALIDATION OF ROOM+EAHE SYSTEM

- It can be observed that the theoretical model is in good agreement with the experimental obtained values.
- Strong correlation coefficients, close to 0.85, is obtained for each day having different air flow velocity.
- The RMS percent deviation is less than 4%, which shows the percent deviation is minimal.
- Hence, the theoretical model results are validated.

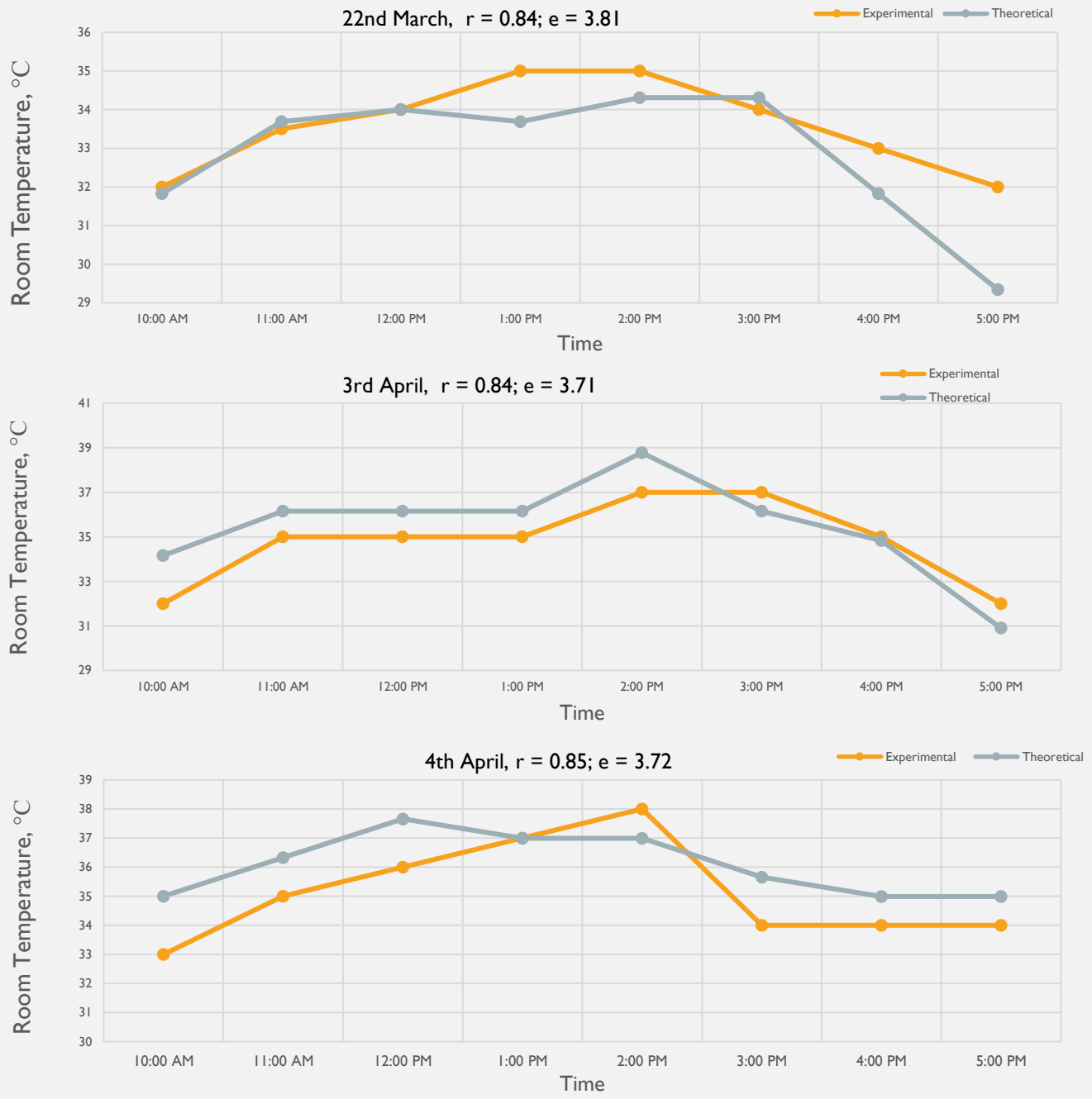


Figure 5 Hourly variation of theoretical room temperature with experimental observation for each day.

REDUCTION IN COOLING LOAD FOR SUMMER SEASON

- The energy consumption of AC without integrating EAHE, running for 8 hours a day, is found to be 14.72 kWh.
- It is found that on an average 479.84 W of cooling load is reduced due to the integration of EAHE. This reduction corresponds to daily saving of 3.8 kWh of power if the system works for 8 hours.
- For the entire summer season, 468 kWh of electricity is saved after integrating EAHE to the room.
- Thus, integrating the EAHE system with AC unit will reduce the cooling load by 20%.

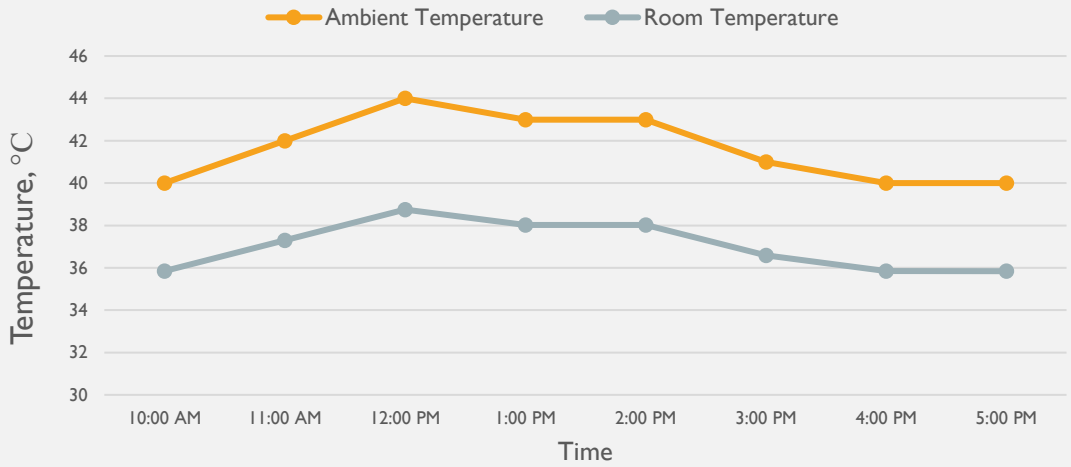


*Figure 6 Hourly variation of ambient temperature and room temperature considering a 12*12*12 ft room.*

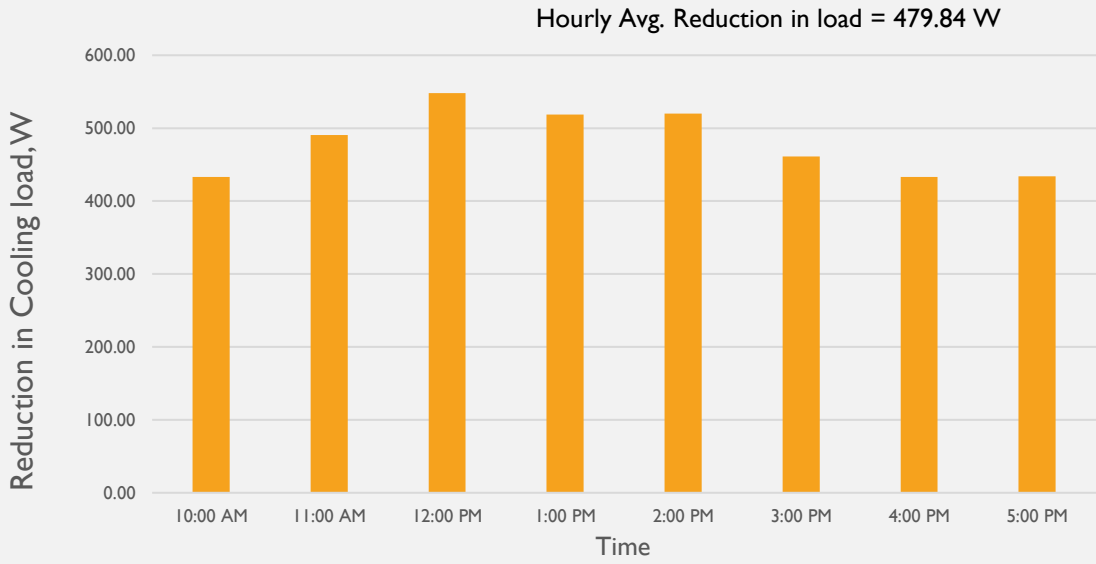


*Figure 7 Hourly reduction in cooling load of the 12*12*12 ft room.*

ENERGY EFFECTIVENESS

	Material Used (in Kg)	Embodied Energy (kWh/kg)	Total embodied energy E_{in} (kWh)
PVC	80	21.11	1681.50
Plastic Fans	1×10	19.44	194.4
Total Embodied Energy			**1875.9**
Annual Energy Output / Year		E_{aout} (kWh)	468
EPBT		4	Years

CARBON CREDIT EARNED

Lifespan of EAHE (Years)	**10**	**15**	**20**	**50**
CO_2 emission (Kg/year)	422.69	281.79	211.34	84.54
CO_2 mitigation (in Ton)	5.32	10.10	14.87	43.51
Carbon Credit Earned (₹)	₹ 4,525	₹ 8,582	₹12,640	₹ 36,986

CONCLUSION

- Temperature difference between the inlet and exit temperature increases as the temperature difference between the ambient air temperature and constant soil temperature increases.
- The validation of the experimental results with both the modelling in MATLAB showed strong correlations, r ranging between 0.82 to 0.89 and low root mean square percent deviation, e ranging between 2.17 to 4.84, confirming both model's accuracy in predicting the exit temperature as well as the room temperature.
- The energy consumption of AC without integrating EAHE, running for 8 hours a day, is found to be 14.72 kWh. The total reduction in cooling load on 4 April for a room of 1.2×12×12 ft is found to be 3.2 kWh and for the entire summer season, it is found to be 468 kWh reduction in cooling load. Thus, integrating the EAHE system with AC unit will reduce the cooling load by 20%.
- The EPBT of the room + EAHE system is found to be 4 years when the EAHE works in summer season. CO_2 mitigation and carbon credit earned for 50 years of EAHE are found to be 43.50 Ton and ₹ 36,986, respectively which contributes to environmental sustainability.

FUTURE WORK



The experimental should be conducted for winter application mostly during December and January.



The total reduction of heating load should be calculated.



This way, total energy saved in heating as well as cooling the room throughout the year can be analysed.

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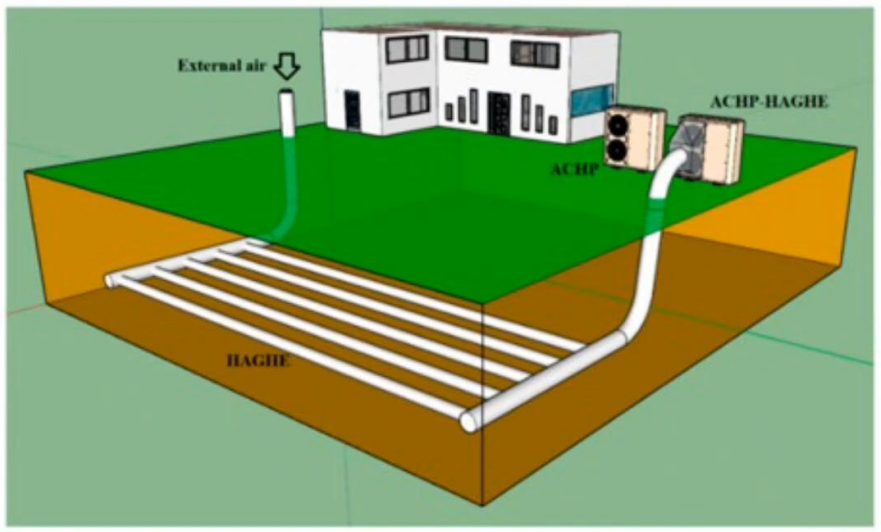
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THANK YOU

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